Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 31)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

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0 1 Table 1 lists some feedback mechanisms that affect the Earth's climate.

0 1 . 1 Complete Table 1 by stating whether each mechanism is an example of **negative** or **positive** feedback.

[4 marks]

Table 1

Mechanism	Type of feedback	
Higher temperatures increase cloud cover, raising albedo		
Melting permafrost releases methane		
Higher CO_2 levels increase photosynthesis and carbon sequestration		
Melting ice reduces albedo		

0 1 . 2 Explain why natural systems regulated by negative feedback rarely reach tipping points.

[2 marks]

0 1 . 3 Using an example, explain how a large unusual event can cause a tipping point.

[2 marks]

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0 1 . 4 Explain how the human response to negative feedback in agriculture can make food production less sustainable.

[2 marks]

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0 2 Table 1 shows the running mean of moth catches as a student increased the number of nights of light-trapping at one site.

Table 1

Number of nights	**1**	**2**	**4**	**6**	**8**	**10**	**12**	**14**
Running mean catch per night	31	18	24	21	22	22	21	22

0 2 . 1 Use Table 1 to suggest the minimum number of nights needed to obtain a reliable mean. Explain your answer.

[2 marks]

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0 2 . 2 Explain why moth surveys must be carried out at standardised times and weather conditions.

[3 marks]

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0 2 . 3 Distinguish between **accuracy** and **precision** of measurements.

[2 marks]

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0 2 . 4 Explain the importance of a null hypothesis and standardised techniques in a scientific investigation.

[3 marks]

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0 3 . 1 Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber .....

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Fuel .....

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Food .....

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Fibres .....

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Medicines .....

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0 3 . 2 Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

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0 3 . 3 Explain how forests regulate the hydrological cycle and local climate.

[2 marks]

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0 4 . 1 Describe how agriculture alters the hydrological cycle.

[3 marks]

0 4 . 2 Explain how each of the following consumer choices can reduce the environmental impact of food production.

[4 marks]

Seasonal / local food

Organic food

0 4 . 3 Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

[3 marks]

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0 5 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 5 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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0 6 The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

0 6 . 1 Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

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Answer [ ] thousand tonnes

0 6 . 2 Calculate the maximum catch that would have kept the stock constant.

[1 mark]

Answer [ ] thousand tonnes

0 6 . 3 Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

[3 marks]

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0 6 . 4 Describe how each of the following measures obtained from commercial catches can indicate overfishing.

[4 marks]

Catch per unit fishing effort .....

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Mean fish size and mean age .....

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0 7 . 1 Define each of the following ecological terms and give an example.

[8 marks]

Species .....

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Ecological niche .....

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Community .....

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Biome .....

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0 7 . 2 Explain the difference between an ecosystem and a biome.

[2 marks]

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0 8 Table 1 shows some features of Earth, Mars and Venus.

Table 1

Feature	Venus	Earth	Mars
Mean distance from Sun / million km	108	150	228
Mass relative to Earth	0.82	1.0	0.11
Atmospheric pressure relative to Earth	92	1.0	0.006
Mean surface temperature / °C	460	15	-60
Magnetic field	Very weak	Strong	Very weak
Rotation period / hours	5 800	24	25

0 8 . 1 Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

[3 marks]

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0 8 . 2 Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

[3 marks]

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0 8 . 3 Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

0 8 . 4 Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

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0 9 . 1 Describe how cloning could be used to conserve the northern white rhinoceros, and explain why success has been limited.

[3 marks]

0 9 . 2 Explain how micro-propagation and cloning are used to conserve rare plants such as the Atlantic black poplar.

[3 marks]

0 9 . 3 Describe the purpose and methods of the Millennium Seed Bank.

[4 marks]

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1 0 . 1 Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

[9 marks]

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1 0 . 2 State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

OR

1 1 . 2 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS